

Multiple Choice (10 marks)

1. When does a buffer overflow occur, generally speaking?
 - (a) when copying a buffer from the stack to the heap
 - (b) when writing to a pointer that has been freed
 - (c) when the program runs out of memory
 - (d) when a pointer is used to access memory not allocated to it
 - (e) when a buffer is not initialized before being used
2. In Message Confidentiality, the transmitted message must make sense to only intended
 - (a) Network Provider
 - (b) Relay station
 - (c) Receiver
 - (d) Sender
 - (e) Interceptor
3. What is the value of this python expression: $4 + 3 \% 2$?
 - (a) 9
 - (b) 7
 - (c) 0
 - (d) 4
 - (e) 8
4. Suppose a convex 3d-object has 15 vertices and 39 edges. How many faces does it have?
 - (a) 25
 - (b) 26
 - (c) 20
 - (d) 31
 - (e) 22
5. Which of the following tasks can be best solved using Clustering.
 - (a) Translating a text from one language to another
 - (b) Detecting fraudulent credit card transactions
 - (c) All of the above
 - (d) Training a self-driving car
 - (e) Predicting the stock market trends

True / False (10 marks)

Answer True or False

6. True or False: Mutation-based fuzzing creates completely new inputs for each test without relying on any previous inputs.
7. True or False: RoBERTa pretrains on a corpus that is significantly larger than the corpus BERT v1.1 used.
8. True or False: Layer Normalization was the primary normalization technique used in the original Transformer paper.
9. True or False: The output of the expression 'a' + 'ab' in Python 3 is 'aab'.
10. True or False: An unconditional branch instruction in APL is specified using a right pointing arrow.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to subtract the elements of the given nested tuples.
12. Write a function to find the sequences of one upper case letter followed by lower case letters.

End of Paper